

M. Ammad Ahmed SE-041 "Calculus"

CEP

"Maclaurin Series"

Problem:

You are analyzing the vibration behavior of mechanical system, specifically a cantilever beam with a concentrated load at its free end. The deflection $y(x)$ of the beam can be approximate using the following equation based on Euler-Bernoulli beam.

$$y(x) = \frac{P}{6EI} (3Lx - x^2)$$

where, • P is the applied load (N)

• E is the young's modulus of the beam

• I is the moment of inertia

• L is the length

• x is the distance from the fixed end of the beam.

Use maclaurin series to analyze deflection.

Objective:

1) Derive the Maclaurin series exp- of $y(x)$ around $x=0$ up to the third degree.

2) Determine the approximate deflection $y(x)$ when

$$P = 1000 \text{ N}, E = 200 \times 10^9 \text{ Pa}, I = 1.67 \times 10^{-8} \text{ m}^4$$

$$L = 2$$

Sol:-

Derive Maclaurin Series expansion of $y(x)$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

1) calculate $y(0)$:

$$y(0) = \frac{P}{6EI} (3L(0) - (0)^2) = 0$$

• calculate first derivative

$$y'(x) = \frac{d}{dx} \left(\frac{P}{6EI} (3L(x) - x^2) \right)$$

$$y'(x) = \frac{P}{6EI} (3L - 2x)$$

• Now calculate $y'(0)$

$$y'(0) = \frac{P}{6EI} [(3L) - 2x]$$

$$= \frac{P}{6EI} (3L)$$

$$y'(0) = \frac{P \cdot L}{6EI}$$

• Calculating 2nd derivative:

$$y''(x) = \frac{d}{dx} \left(\frac{P}{6EI} (3L - 2x) \right)$$

$$y''(x) = \frac{P}{6EI} (-2) = -\frac{P}{3EI}$$

* $y''(0)$ ~ calculating

$$y''(0) = -\frac{P}{3EI}$$

• Calculate Third derivative:

$$y'''(x) = 0 \quad \text{Since 2nd derivative is constant.}$$

Now substituting these into Maclaurin series.

$$y(x) = 0 + \frac{P \cdot L}{2EI} (x) + \frac{-P}{3EI} \frac{x^2}{2} + 0$$

Simplifying gives:

$$y(x) = \frac{P \cdot L}{2EI} (x) - \frac{P}{6EI} (x^2)$$

Step 2:

Approximate deflection for given values:

substituting these value into deflection eq:

• Calculate $\frac{P \cdot L}{2EI}$:

$$\frac{P \cdot L}{2 \cdot EI} = \frac{1000 \cdot 2}{2 \cdot (200 \times 10^9) (1.67 \times 10^{-8})}$$

$$\frac{P \cdot L}{2 \cdot EI} = 299.70 \text{ m}^{-1}$$

• Calculate $\frac{-P/3EI}{2}$:

$$\frac{-P/3EI}{2} = \frac{\frac{-1000}{3 \cdot (200 \times 10^9) (1.67 \times 10^{-8})}}{2}$$

$$\frac{-P/3EI}{2} = -49.87 \text{ m}^{-2}$$

Results:

- The first term $299.70x$ indicates a linear relationship b/w the deflection and the distance from fixed end. As x increases, the deflection increase
- The second term $-49.87x^2$ shows that the deflection curve will eventually decrease due to quadratic term, which is expected as cantilever beam starts bending downward under the load.

Conclusion:

The use of Maclaurin Series provides a powerful tool to approximate the deflection of the cantilever beam under load, helping in the design and analysis of structural components in engineering problem.

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'Masab Ahmed'

'SE-043'

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PROBLEM:- A company is designing a closed cylindrical tank with a fixed volume of 500 m^3 . The tank consists of a circular base, a circular top and a side wall. To minimize the material cost, the goal is to determine the dimensions (radius and height) that minimize the surface area while maintaining the fixed volume.

OBJECTIVE:- Find the radius (r) and height (h) of the cylinder that minimize the surface area, subject to the volume constraint.

FORMULATION:-

VOLUME CONSTRAINT:-

The volume (V) of the cylinder is,

$$V = \pi r^2 h = 500 \text{ m}^3$$

From this, we can express height (h) in terms of radius (r).

$$h = \frac{500}{\pi r^2}$$

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2. SURFACE AREA:-

The surface area (A) (including top and bottom) is given,

$$A = 2\pi r^2 + 2\pi rh$$

Substituting 'h' from volume constraint

$$A(r) = 2\pi r^2 + 2\pi r \left(\frac{500}{\pi r^2} \right) = 2\pi r^2 + \frac{1000}{r}$$

3. OPTIMIZATION:-

To minimize the surface area we differentiate $A(r)$ with respect to r and set it equal to zero to find the critical points.

First compute $\frac{dA}{dr}$

$$\frac{dA}{dr} = 4\pi r - \frac{1000}{r^2}$$

4. CRITICAL POINTS:-

Set $\frac{dA}{dr} = 0$ to find critical points

$$4\pi r = \frac{1000}{r^2}$$

Multiply both sides by r^2

$$4\pi r^3 = 1000$$

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Solving for r ,

$$r^3 - 1000 = r^2 \cdot 3 \left| \begin{array}{l} 250 \\ \pi \end{array} \right.$$

Approximate value,

$$r = 4.301 \text{ m}$$

HEIGHT:-

Now, substitute r back into the volume equation to find h ,

$$h = \frac{500}{\pi (4.301)^2}$$

$$h = 8.601 \text{ m}$$

CONCLUSION:-

The radius and height that minimize the surface area while maintaining the fixed volume are approximately $r = 4.301 \text{ m}$ and $h = 8.601 \text{ m}$.

Q A Landscape architect is designing a park and wants to maximize the area of a flower bed. The area function can be modelled as

$$A(x, y) = 100x + 150y - 4x^2 - 3y^2$$

where x is the length and y is the width of the flower bed.

- Find critical points of area function.
- Determine whether these critical points correspond to maximum, minimum or saddle point.
- Explain how results can help the architect in designing the flower bed.

$$p \frac{\partial A}{\partial x} = 100 + 0 - 8x = 100 - 8x$$

$$q \frac{\partial A}{\partial y} = 150 - 0 - 6y = 150 - 6y$$

$$r \frac{\partial^2 A}{\partial x^2} = -8$$

$$t \frac{\partial^2 A}{\partial y^2} = -6$$

$$s \frac{\partial^2 A}{\partial x \partial y} = 0$$

$$\begin{array}{l|l} \frac{\partial A}{\partial x} = 100 - 8x & \frac{\partial A}{\partial y} = 150 - 6y \\ 0 = 100 - 8x & 0 = 150 - 6y \\ 100/8 = x & 150/6 = y \\ 25/2 = x & 25 = y \end{array}$$

$$\text{For } (a, b) \rightarrow (25/2, 25)$$

$$r = -8$$

$$s = 0$$

$$t = -6$$

$$rt - s^2 \rightarrow -8(-6) - 0^2 \rightarrow 48$$

as $rt - s^2 > 0$ and $r < 0$,

then $A(x, y)$ has maximum value at critical points.

Conclusion:

- At $x = 25/2$ and $y = 25$, the area of flower bed will be maximized.

Application:

In many real world problems we want to maximize or minimize a certain quantity for optimization hence we use maximum and minimum values for this purpose.

In engineering, this concept is used to design structures, optimizing materials and ensuring safety.

In the field of technology especially in machine learning optimization techniques are used to minimize error in predictive models.

3/Oct/24

COMPLEX ENGINEERING PROBLEM

Q- A company has a cash flow function given by $C(t) = 150t^3 - 30t^2 + 500$, where t is the time in months. The company want to determine the time when the rate of change of cash flow is a minimum, indicating a potential turning point in their financial stability.

Solution:

$$C(t) = 150t^3 - 30t^2 + 500$$

First Derivative:

$$C'(t) = (3)150t^2 - (2)30t + 0$$

$$C'(t) = 450t^2 - 60t$$

$$\text{for } t, \quad C'(t) = 0$$

$$0 = 450t^2 - 60t$$

$$0 = (15t - 2)30t$$

hence,

$$30t = 0$$

$$t = 0$$

and,

$$15t - 2 = 0$$

$$t = \frac{2}{15}$$

Second derivative:

$$C''(t) = 900t - 60$$

for $t = 0$

$$C''(0) = 0 - 60$$

$$C''(0) = -60$$

$$\therefore C''(0) < 0$$

maxima

for $t = \frac{2}{15}$

$$C''\left(\frac{2}{15}\right) = 900 \times \frac{2}{15} - 60$$

$$C''\left(\frac{2}{15}\right) = 120 - 60$$

$$C''\left(\frac{2}{15}\right) = 60$$

$$\therefore C''\left(\frac{2}{15}\right) > 0$$

minima

Result :

Therefore, the time at which the rate of change of the cash flow is at a minimum is $t = \frac{2}{15}$ months or 0.133 months.

COMPLEX ENGINEERING PROBLEM

BACKGROUND:-

When planning a garden, one often considers not only the total area that can be enclosed with a given amount of fencing but also how to maximize this area efficiently. In this case, we have 100 meters of fencing available to enclose a rectangular garden. To enhance functionality, we want to divide the garden into two equal rectangular sections by adding a fence parallel to one of the garden's sides. The challenge is to determine the optimal dimensions that will maximize the area of garden while satisfying the fencing constraints.

FORMULA:

Let x be the length of garden

Let y be the width of garden.

The total length of fencing will be given by perimeter, including the additional dividing fence.

The total length of the fencing can be expressed as:

$$P = 2x + 2y + y = 100$$

This simplifies to

$$P = 2x + 3y = 100$$

From this equation, we can express y in terms of x :

$$3y = 100 - 2x$$

$$y = \frac{100 - 2x}{3}$$

The Area " A " of the garden can be expressed as:

$$A = x \cdot y$$

$$A = x \left(\frac{100 - 2x}{3} \right)$$

This simplifies to:

$$A = \frac{100x - 2x^2}{3}$$

To find maximum area, we will differentiate A w.r.t

"x" and set the derivative to zero.

SOLUTION:

Differentiate the area function:

$$A = \frac{100x - 2x^2}{3}$$

Differentiate w.r.t "x"

$$\frac{dA}{dx} = \frac{d}{dx} \left(\frac{100x - 2x^2}{3} \right)$$

$$\frac{dA}{dx} = \frac{100 - 4x}{3}$$

Now, Set the derivative to zero: To find the critical point

$$\frac{100 - 4x}{3} = 0$$

$$100 - 4x = 0$$

$$+ 4x = +100$$

$$x = \frac{100}{4}$$

$$x = 25$$

Calculate y: Substitute x back into the equation for

$$y = \frac{100 - 2(25)}{3}$$

$$y = \frac{100 - 50}{3}$$

$$y = \frac{50}{3}$$

$$y = 16.67$$

Check for maximum: To ensure this is a maximum,
Check the Second derivative:

$$\frac{d^2A}{dx^2} = -\frac{4}{3}$$

Since, the Second derivative is negative, this confirms a local maximum.

RESULTS:

The dimensions that maximize the total area of the garden are:

$$\text{Length } (x) = 25 \text{ m}$$

$$\text{Width } (y) = 16.67 \text{ m}$$

The total area is:

$$A = x \cdot y$$

$$A = 25 \times 16.67$$

$$A = 416.67 \text{ m}^2$$

INTERPRETATION:

By using 100 m of fencing to enclose the rectangular garden and dividing it into two equal sections, the optimal dimensions are 25m for the length and approximately 16.67m for the width. This configuration results in maximum area of approximately 416.67m². The division allows for better organization and usability of the garden space, meeting both the area maximization objective and functional needs. This approach demonstrates how strategic planning can lead to optimal use of resources in garden design.

TOPIC:- Optimization of Rectangular Box design using partial differentiation

Finding:- An Engineering team is tasked with designing a rectangular box for packaging a product. The goal is to minimize the amount of material used to construct the box while ensuring it can hold a specific volume of 500 cubic meters.

Formulation:-

The Volume of the box is given by,

$$V = L \times W \times H = 500 \text{ m}^3$$

The Surface area S of the box is expressed as,

$$S = 2(LW + LH + WH)$$

from the Volume constraint, we can express height in terms of length and width,

$$h = \frac{500}{LW}$$

Substituting it into surface area equation.

$$S = 2 \left[LW + L \left(\frac{500}{LW} \right) + W \left(\frac{500}{LW} \right) \right]$$

Simplifying,

$$S = 2 \left(LW + \frac{500}{W} + \frac{500}{L} \right)$$

Solution:-

To find critical points, we have to minimize S by taking partial derivatives w.r.t L & W and set them to zero.

$\Rightarrow$ partial derivative w.r.t L ,

$$\frac{\partial S}{\partial L} = 2(W + 0 - 500L^{-2}) = 0$$

$$\frac{\partial S}{\partial L} = 2\left(W - \frac{500}{L^2}\right) = 0$$

$$W = \frac{500}{L^2}$$

$\Rightarrow$ partial derivative w.r.t W ,

$$\frac{\partial S}{\partial W} = 2\left(L - 500W^{-2} + 0\right) = 0$$

$$\frac{\partial S}{\partial W} = 2\left(L - \frac{500}{W^2}\right) = 0$$

$$L = \frac{500}{W^2}$$

Substituting $W = 500/L^2$ into $L = 500/W^2$

$$L = \frac{500}{\left(\frac{500}{L^2}\right)^2}$$

$$L^3 = 500$$

$$L = \sqrt[3]{500} \Rightarrow L^2 = \left(\sqrt[3]{500}\right)^2 \Rightarrow L^2 = 500^{2/3}$$

Now compute W & H .

$$W = \frac{500}{500^{2/3}}$$

$$W = (500)^{1-2/3}$$

$$W = 500^{1/3}$$

$$W = \sqrt[3]{500}$$

$$H = \frac{500}{(3\sqrt{500})(3\sqrt{500})}$$

$$H = \frac{500}{(3\sqrt{500})^2}$$

$$H = \frac{500}{500^{2/3}}$$

$$H = 500^{1-2/3}$$

$$H = 500^{1/3}$$

$$H = 3\sqrt[3]{500}$$

Result:-

The optimal dimension's that minimize the surface area while maintaining a volume of 500m^3 are

$$\text{Length} = L = 3\sqrt[3]{500} \approx 7.937\text{m}$$

$$\text{Width} = W = 3\sqrt[3]{500} \approx 7.937\text{m}$$

$$\text{Height} = H = 3\sqrt[3]{500} \approx 7.937\text{m}$$

Conclusion:-

The analysis demonstrates that to minimize the surface area of a rectangular box while maintaining a fixed volume of 500m^3 , the optimal dimensions should be equal. The resulting dimensions of

approximately 7.937 meters for length, width and

height ^{indicate} that the most efficient design is a cube.

This outcome not only illustrates the effectiveness of mathematical optimization techniques but also emphasizes the benefit of resource conservation and sustainability design.

Interpretation:-

The design of a rectangular box with a specified volume of 500m^3 exemplifies the intersection of engineering and optimization. By focusing on minimizing surface area while maintaining volume, the team aims to reduce material cost and environmental impact. This optimization not only enhances efficiency of the packaging process but also ensure product integrity during and storage.

COMPLEX ENGINEERING PROBLEM

Date: _____

BACKGROUND

- A small aircraft manufacturer hires you to design a fuel tank that optimizes metal usage while meeting specific design requirements. The tank's construction involves a rectangular piece for the lateral surface and square pieces for top and bottom.

PROBLEM STATEMENT

- Design a cylindrical fuel tank with minimum metal usage, considering the volume requirements and material constraints.

OBJECTIVE

- Minimize the total surface area of the fuel tank to reduce metal usage. Determine dimensions as well.

DESIGN REQUIREMENTS

- Volume (V): 1000 inches^3
- Lateral surface material: Aluminium alloy
- Top and bottom material: Aluminium alloy (stamped from square pieces)
- Minimize metal usage

CONSTRAINTS

- Tank height (h) > 0
- Square pieces' waste consideration
- Material thickness = const.
- Tank radius (r) > 0

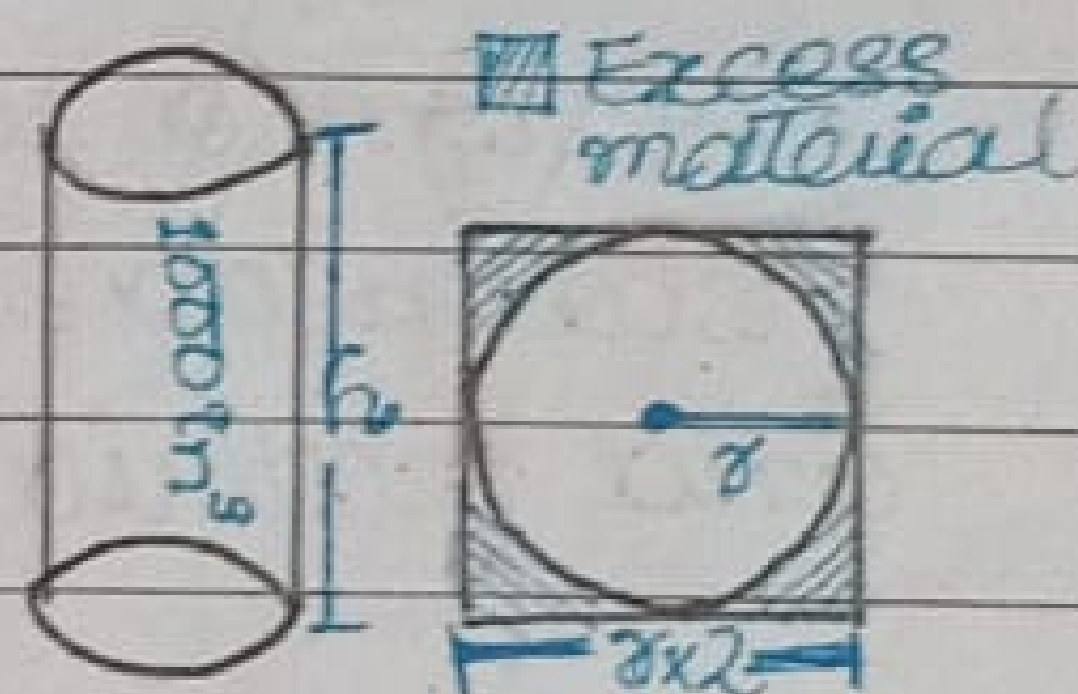
Solution

- The material used in the construction of the fuel tank uses three pieces. The surface area of the lateral surface of the tank is $2\pi rh$. The area of the square pieces used for the construction of top and bottom is $(2r)^2$. (We must take into account the wasted material as well as the material used for the circular end)
- Hence, the total surface area can be given as follows:

$$S = 2\pi rh + (2r)^2 + (2r)^2$$

$$\Rightarrow S = 2\pi rh + 4r^2 + 4r^2$$

$$\Rightarrow S = 2\pi rh + 8r^2$$



$$S = 2\pi rh + 8r^2 \rightarrow (i)$$

- Since the volume of the fuel tank must be 1000 inches³, we can write:

$$V = \pi r^2 h$$

$$\Rightarrow 1000 = \pi r^2 h$$

$$\Rightarrow 1000 / \pi r^2 = h$$

$$\Rightarrow h = 1000 / \pi r^2$$

$$h = 1000 / \pi r^2 \rightarrow (ii)$$

- Substituting the value of 'h' from (ii) in (i):

$$S(r) = 2\pi r (1000 / \pi r^2) + 8r^2$$

$$\Rightarrow S(r) = 2000 / r + 8r^2$$

$$\Rightarrow S(r) = 2000r^{-1} + 8r^2$$

$$S(x) = 2000x^{-1} + 8x^2 \rightarrow \text{iii)}$$

• To determine the minimum surface area we must first find the critical points of $S(x)$, which can be done as follows:

• differentiating $S(x)$ with respect to x :

$$\begin{aligned} \bullet S'(x) &= d/dx (2000x^{-1} + 8x^2) \\ \Rightarrow S'(x) &= (2000)(-1)(x)^{-1-1} + (8)(2)(x)^{2-1} \\ \Rightarrow S'(x) &= -2000x^{-2} + 16x \end{aligned}$$

• Putting $S'(x) = 0$ and solving for x :

$$\begin{aligned} \bullet -2000x^{-2} + 16x &= 0 \\ \Rightarrow -2000/x^2 &= -16x \\ \Rightarrow 2000 &= 16x(x^2) \\ \Rightarrow 16x^3 &= 2000 \\ \Rightarrow x^3 &= 125 \end{aligned}$$

• Taking cube root on both sides:

$$\begin{aligned} \bullet \sqrt[3]{x^3} &= \sqrt[3]{125} \\ \Rightarrow x &= 5 \end{aligned}$$

$$x = 5$$

• Now, we must determine if it is a point of maxima or minima, this can be done as follows:

• Putting $x = 5-h$ i.e. $x = 5-0.1$ or $x = 4.9$ in $S'(x)$:

$$\begin{aligned} \bullet S'(4.9) &= -2000(4.9)^{-2} + 16(4.9) \\ \Rightarrow S'(4.9) &= -83.2986 + 78.4 \\ \Rightarrow S'(4.9) &= -4.8986 \end{aligned}$$

$$S'(4.9) = -4.8986$$

• Putting $r = 5 + h$, i.e. $r = 5 + 0.1$ or $r = 5.1$ in $S'(r)$:

$$\begin{aligned} \bullet S'(5.1) &= -2000(5.1)^{-2} + 16(5.1) \\ \Rightarrow S'(5.1) &= -76.8935 + 81.6 \\ \Rightarrow S'(5.1) &= 4.7065 \end{aligned}$$

$$S'(5.1) = 4.7065$$

• Therefore, it can be deduced that:

- ♥ $S'(5+0.1) > 0$ and $S'(5-0.1) < 0$
- ♥ Surface area is minimum at $r = 5$

• So, the surface area is minimized when the radius is 5. However, we still need to determine the height, therefore putting $r = 5$ in (ii):

$$\begin{aligned} \bullet h &= 1000/\pi(5)^2 \\ \Rightarrow h &= 40/\pi \\ \Rightarrow h &= 12.732 \end{aligned}$$

$$h = 12.732$$

• The wasted material will be:

$$\bullet W = 2 \times (\text{Area of square} - \text{Area of circle})$$

$$\Rightarrow W = 2 \times (15 \times 2)^2 - (\pi \times 5^2)$$

$$\Rightarrow W = 2 \times (100 - 78.539)$$

$$\Rightarrow W = 2 \times (21.46)$$

$$\Rightarrow W = 42.92 \text{ inches}^2$$

- So, the Total waste material from top and bottom circular portions of the fuel tank is 42.92 inches^2 .

$$W = 42.92$$

- The length of the side of the square aluminium alloy pieces will be:

- $\text{length} = 2 \times r$

$$\Rightarrow \text{length} = 2 \times 5$$

$$\Rightarrow \text{length} = 10 \text{ inches}$$

$$\text{length of side of square} = 10 \text{ inches}$$

- The total surface area of all the materials used will be:

- $S_m = (2000)(5)^2 + 8(5)^2$

$$\Rightarrow S_m = 2000(5) + 8(25)$$

$$\Rightarrow S_m = 400 + 200$$

$$\Rightarrow S_m = 600 \text{ inches}^2$$

$$\text{minimum surface area} = 600 \text{ inches}^2$$

- And the surface area of the cylinder will be:

$$S_c = S_m \quad \bullet \quad S_c = S_m - W$$

$\Rightarrow S_c = 600 - 42.92$

$\Rightarrow S_c = 557.08 \text{ inches}^2$

Surface area of cylinder = 557.08 inches^2

• RESULT:

• It is found that the optimal dimensions are:

• Radius (r) = 5 inches

• Height (h) = 12.732 inches

• Therefore, the length of each side of square for top and bottom is 10 inches.

• The surface area of the fuel tank is 557.08 inches^2

• The total area of aluminium alloy needed is 600 inches^2 and out of this 42.92 inches^2 will be discarded.

• CONCLUSION:

• The cylindrical tank can be constructed efficiently with minimal material usage while fulfilling the volume requirement of 1000 inches^3 . The design takes into account the dimensions that optimize the use of aluminium alloy particularly for the lateral surface and stamped square pieces.

• INTERPRETATION:

• The results indicate a well balanced tank design where radius and height are optimized for reduced surface area, thus minimizing material usage and cost.

Date:

The calculated total surface area reflects the effective use of the aluminum alloy while ensuring structural integrity.

APPLICATION

This design is directly applicable in the aircraft manufacturing industry, where fuel efficiency and material costs are critical. Implementing this optimized cylindrical tank can lead to a lighter aircraft, improved fuel economy and reduced manufacturing expenses.

Additionally, the principles used in this design can be applied to other cylindrical storage tanks in various industries, such as oil, water, chemical substances and other materials, where volume constraints and material efficiency are significant considerations.

PROBLEM:-

In a design of an aerospace wing surface, engineers are analyzing the deformation properties under a combined aerodynamic load. The deformation is represented by the function.

$$f(x, y) = x^3 + x^2y^2 + 2y^3 + 2x + y$$

where 'x' and 'y' represent spatial coordinates on the surface, and $f(x)$ describes the deflection of the surface at each point. The goal is to "analyze the curvature" of the wing surface in the 'x' and 'y' directions to understand the behaviour of stresses and deflections at various points.

Task:

- 1- Compute the 2nd derivatives $\frac{\partial^2 f}{\partial x^2}$ & $\frac{\partial^2 f}{\partial y^2}$, which represent the curvature of the wing surface in 'x' and 'y' directions respectively.
- 2- Use this curvature information to evaluate the points of maximum stress and potential failure zones on the wings.

Solution:

$$f(x, y) = x^3 + x^2y^2 + 2y^3 + 2x + y$$

First Partial Derivatives:

w.r.t x :

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^3 + x^2y^2 + 2y^3 + 2x + y)$$

$$\frac{\partial f}{\partial x} = 3x^2 + 2xy^2 + 2$$

w.r.t. y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} (x^3 + x^2y^2 + 2y^3 + 2x + y)$$

$$\frac{\partial f}{\partial y} = 2x^2y + 6y^2 + 1$$

Second Partial Derivatives:

w.r.t. $x \left(\frac{\partial^2 f}{\partial x^2} \right)$:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} (3x^2 + 2xy^2 + 2)$$

$$\frac{\partial^2 f}{\partial x^2} = 6x + 2y^2$$

w.r.t. $y \left(\frac{\partial^2 f}{\partial y^2} \right)$:

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} (2x^2y + 6y^2 + 1)$$

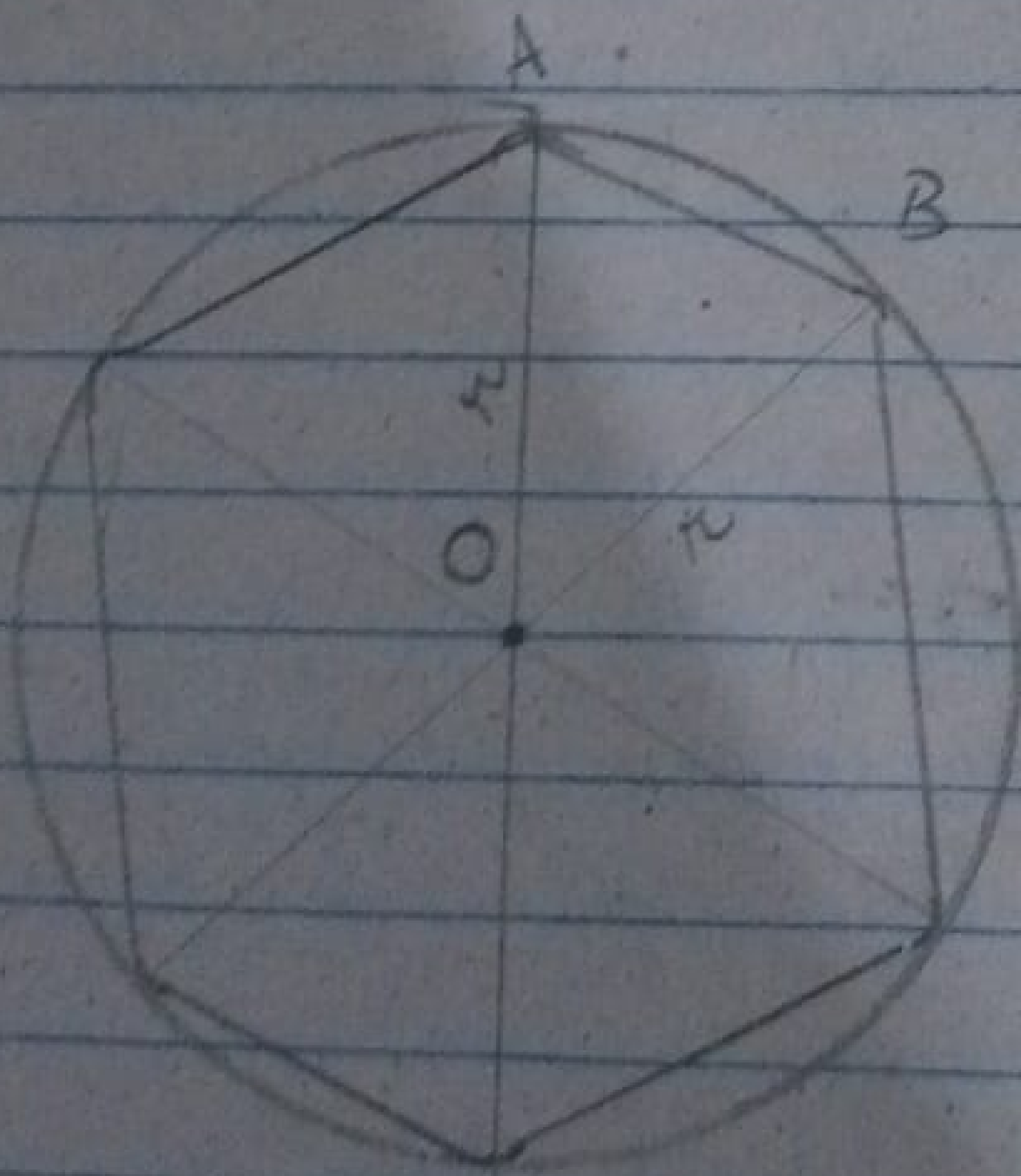
$$\frac{\partial^2 f}{\partial y^2} = 2x^2 + 12y$$

Conclusion:

These curvatures can be used to predict regions of potential failure, as areas with higher curvature are more likely to experience stress concentrations. This info is critical for ensuring the safety and integrity of the wing structure, especially in high-stress flight conditions.

COMPLEX ENGINEERING PROBLEM

PROBLEM STATEMENT:
AREA OF AN
INSCRIBED POLYGON AS
A FUNCTION OF SIDES
AND RADIUS



OBJECTIVE:- Analyze the behaviour of A_n of a regular n -sided polygon inscribed in a circle as a function of the number of sides n and the radius r . Investigate how the area changes with respect to both parameters and determine conditions under which the area maximizes.

GIVEN:- DEFINITION:- A regular n -sided polygon is a polygon with n -equal sides and angles and it can be inscribed in a circle.

FORMULA:- The area A_n of a regular n -sided polygon inscribed in a circle of radius r is given by:

$$A_n = \frac{1}{2} n r^2 \sin\left(\frac{2\pi}{n}\right)$$

COMPONENTS:-

n : Number of sides (vertices) of the polygon.

r : Radius of the circumscribed circle

$\sin\left(\frac{2\pi}{n}\right)$: Sine of the central angle subtended by two adjacent vertices.

DERIVATION:-

CENTRAL ANGLES:- Each vertex of the polygon subtends an angle of $\frac{2\pi}{n}$ at the center of the circle.

TRIANGLES FORMATION:- The polygon can be divided into n -isosceles triangles, each having a vertex at the center and the other two vertices on the circle.

AREA OF ONE TRIANGLE:-

The area of triangle is given by:

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

- Consider two adjacent vertices A and B of the polygon at center O of the circle
- The lengths of the sides OA and OB are both equal to the radius r
- The angle $\angle AOB = \frac{2\pi}{n}$

$$A = \frac{1}{2} \times OA \times OB \times \sin(\angle AOB)$$

$$A = \frac{1}{2} \times r \times r \times \sin\left(\frac{2\pi}{n}\right)$$

$$A = \frac{1}{2} r^2 \sin\left(\frac{2\pi}{n}\right)$$

$$A_n = n \times A = \frac{1}{2} n r^2 \sin\left(\frac{2\pi}{n}\right)$$

What number does A_n approaches when n gets larger?

As n approaches infinity, the area A_n of regular n-sided polygon inscribed in a circle of radius r approaches the area of circle itself. The area of a circle is given by the formula

$$A_{\text{circle}} = \pi r^2$$

To see this mathematically, consider the area formula for the polygon:

$$A_n = \frac{1}{2} n r^2 \sin\left(\frac{2\pi}{n}\right)$$

As n increases, the angle $\frac{2\pi}{n}$ approaches zero and we can use the small angle approximation $\sin(x) \approx x$ where x is near 0. Therefore:

Substituting this approximation into the area formula gives:

$$A_n \approx \frac{1}{2} n r^2 \left(\frac{2\pi}{n} \right) = \pi r^2$$

Thus as n approaches infinity the area A_n approaches:

$$\lim_{n \rightarrow \infty} A_n = \pi r^2$$

In conclusion, the area of the inscribed polygon approaches the area of circle.

Name: Mubashir

Section: A Roll No: 44

Date: _____

Complex Problem:

A manufacturer wants to design an open-top rectangular box with a square base. The box must have a volume of 2 cubic meters. The material for the base costs \$2 per square meter, and the material for the sides costs \$1 per square meter. Find the dimensions of the box that minimize the cost of materials.

Sol Let x be the length of a side of the square base, and h be the height of the box. The volume V of the box is given by:

$$V = x^2 h$$

Given that $V = 2$

$$x^2 h = 2 \Rightarrow \boxed{h = \frac{2}{x^2}} \quad (i)$$

The cost consists of the base and the four sides. The area of the base is x^2 , and the area of the four sides is $4xh$. Therefore, the total cost C is:

$$C = 2 \cdot x^2 + 1 \cdot 4xh$$

Substituting h from the eq (i)

$$\therefore \Rightarrow (i) \quad h = \frac{2}{x^2}$$

$$C = 2x^2 + 4x \left(\frac{2}{x^2} \right)$$

$$\boxed{C = 2x^2 + \frac{8}{x}}$$

To Find the minimum, we take the derivative of C with respect to x and set it to zero:

$$\frac{dC}{dx} = 4x - \frac{8}{x^2}$$

Let,

$$\frac{dC}{dx} = 0$$

So,

$$4x - \frac{8}{x^2} = 0$$

$$4x = \frac{8}{x^2}$$

$$x^3 = 2$$

$$\boxed{x = \sqrt[3]{2}}$$

Substitute it in eq (i)

$$h = \frac{2}{x^2}$$

$$h = \frac{2}{((2)^{\frac{1}{3}})^2}$$

$$h = \frac{2}{(2)^{2/3}}$$

$$h = 2^{1-2/3}$$

$$\boxed{h = 2^{1/3}}$$

We got,

$$x = \sqrt[3]{2} \text{ meters} = 1.26 \text{ meters}$$

$$h = \sqrt[3]{2} \text{ meters} = 1.26 \text{ meters}$$

Conclusion:

The dimensions of the box that minimize the cost of materials are:

- Side length of square base $x \approx 1.26 \text{ meters}$.
- Height $h \approx 1.26 \text{ meters}$

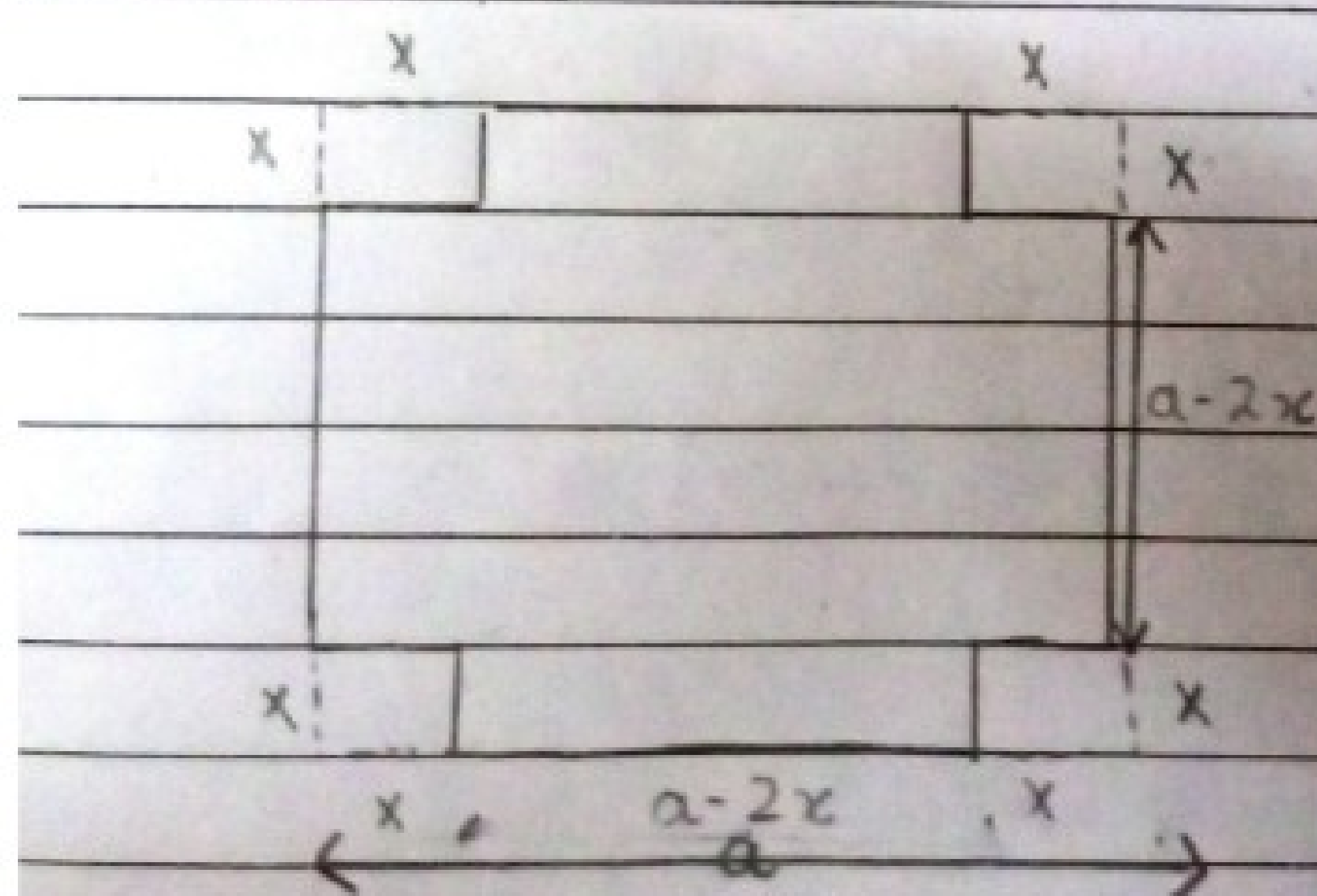
Date: _____

COMPLEX ENGINEERING PROBLEM

STATEMENT

A square sheet of cardboard with each side a centimeter is to be used to make an open-top box by cutting a small square of cardboard from each of the four corners and bending up the sides. What is the side length of the small squares if the box is to have as large a volume as possible?

SOLUTION



Let, the side length of the small square be x cm. The side length of the open box is $(a-2x)$ cm and the height is x cm. Here, a is constant and x is the variable.

The volume V cm³ of the box is given by

$$V(x) = x(a-2x)^2 = 4x^3 - 4ax^2 + a^2x$$

Maximizing the Volume

Differentiate the volume function

$$\frac{dV}{dx} = 12x^2 - 8ax + a^2$$

Setting the Derivative to Zero And Solve for

$$\frac{dV}{dx} = a^2 - 8ax + 12x^2$$

$$a^2 - 8ax + 12x^2 = 0$$

$$x = \frac{-(-8a) \pm \sqrt{(-8a)^2 - 4(12)a^2}}{2(12)}$$

$$x = \frac{8a \pm \sqrt{64a^2 - 48a^2}}{24}$$

$$x = \frac{8a \pm \sqrt{16a^2}}{24}$$

$$x = \frac{8a + 4a}{24}$$

$$x = \frac{12a}{24}$$

$$x = \frac{4a}{24}$$

$$x = \frac{a}{2}$$

$$x = \frac{a}{6}$$

determining which value of x maximizes

we will use second derivative test to figure out

$$\frac{dV}{dx} = 24x - 8a$$

$$x = a/6$$

$$\frac{dV}{dx} = 24\left(\frac{a}{6}\right) - 8a$$

$$\frac{dV}{dx} = 4a - 8a$$

$$\frac{dV}{dx} = -4a$$

Since, $-4a < 0$, which means that

$x = a/6$ is maximum point. Therefore, the length $x = a/6$ will give maximum volume

At $x = a/2$

If $x = a/2$, the length of each side of the box (after cutting out the corners) becomes;

$$a - 2x \Rightarrow a - 2\left(\frac{a}{2}\right) \Rightarrow a - a \Rightarrow 0$$

This means the box has no base area because the length of each side is zero which makes volume zero.

Thus, when $x = a/2$, there is no actual box.

Maximum Volume Calculation

Now, we know the max. volume occurs at $x = a/6$; substitute into volume formula

$$V(x) = x(a - 2x)^2$$

$$V\left(\frac{a}{6}\right) = \frac{a}{6} \left(a - 2 \times \frac{a}{6}\right)^2$$

$$V\left(\frac{a}{6}\right) = \frac{a}{6} \left(\frac{2a}{3}\right)^2$$

$$V\left(\frac{a}{6}\right) = \frac{a}{6} \left(\frac{4a^2}{9}\right)$$

$$V\left(\frac{a}{6}\right) = \frac{2a^3}{27}$$

So, the max. volume is $V\left(\frac{a}{6}\right) = \frac{2a^3}{27}$



CONCLUSION

The maximum volume occurs when $x = \frac{a}{6}$

and the maximum volume is $\frac{2a^3}{27}$ so

the easiest way to solve this problem is to cut the small squares of size $x = a/6$ from each corner. This will give you an open-top box with maximum volume possible.

REAL WORLD USE

Consider you are a packaging company and you need to create a box using the least amount of material while still maximizing its capacity. This could be useful in minimizing waste, lowering cost and maximizing storage space.

FIELD OF THE PROBLEM

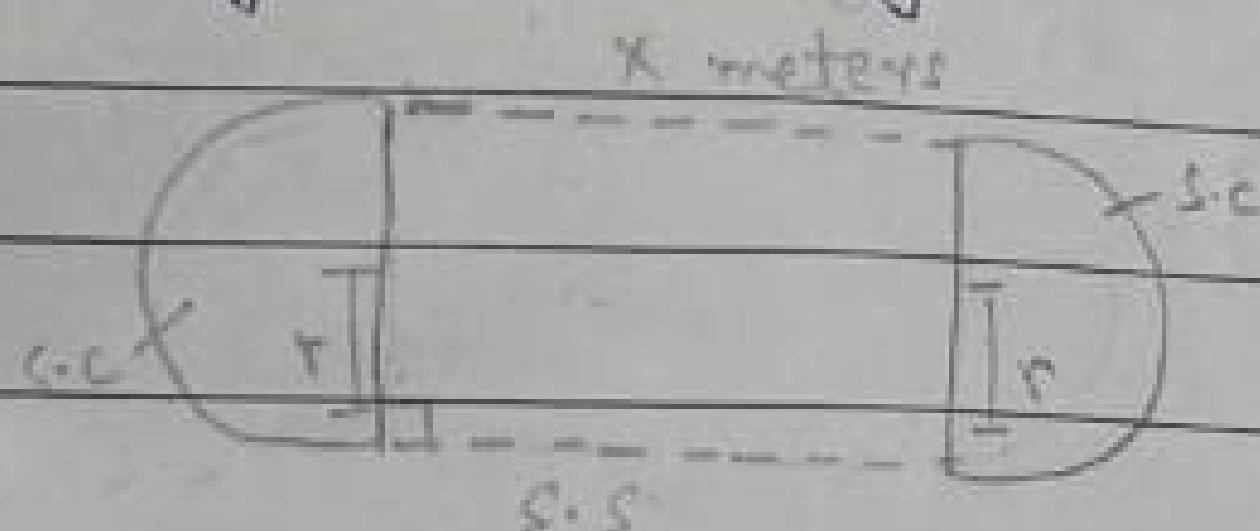
The problem uses differentiation and maximization methods from Calculus and applies them to real world fields such as manufacturing and engineering design. This shows how Calculus has a key role in solving practical problems.

Date: _____

The inside lane of a school running track consists of two straight sections, each of length x meters, and two semicircular sections each of radius r meters. The straight sections are perpendicular to the diameters of the semicircular sections. The perimeter of the inside lane is 400 meters.

- (i) Show that the area, $A \text{ m}^2$, of the region enclosed by the inside lane is given by $A = 400r - \pi r^2$.
- (ii) Given that x and r can vary, show that, when A has a stationary point, there are no straight sections in the track. Determine whether the stationary value is maximum or minimum.

Drawing the diagram



(i) Deriving an equation for Area (A)

$$\text{Perimeter} = \frac{2\pi r}{2} + \frac{2\pi r}{2} + x + x$$

$$= 2\pi r + 2x$$

$$400 = 2(\pi r + x)$$

$$200 = \pi r + x$$

$$x = 200 - \pi r \quad \text{--- eq (1)}$$

$$A = \frac{\pi r^2}{2} + \frac{\pi r^2}{2} + [2x(200 - \pi r)]$$

$$= \pi r^2 + [400r - 2\pi r^2]$$

$$A = \pi r^2 + 400r - 2\pi r^2$$

$$A = 400r - \pi r^2$$

$$(ii) A = 400r - \pi r^2$$

$$A' = 400 - 2\pi r$$

$$400 - 2\pi r = 0$$

$$400 = 2\pi r$$

$$200 = \pi r$$

$$\boxed{\frac{200}{\pi} = r}$$

$$X = 200 - \pi r$$

$$= 200 - \pi \left(\frac{200}{\pi} \right)$$

$$\boxed{X = 0}$$

$$\text{At } r = \frac{200}{\pi}$$

$$\text{Put } r = \frac{200}{\pi} - h, \text{ i.e. } r = \frac{200}{\pi} - 0.1 = 63.56$$

$$f'(63.56) = 400 - 2\pi(63.56)$$

$$\boxed{= 0.6283 +ve}$$

$$\text{Put } r = \frac{200}{\pi} + h, \text{ i.e. } r = \frac{200}{\pi} + 0.1 = 63.762$$

$$f'(63.762) = 400 - 2\pi(63.762)$$

$$\boxed{= -0.6283 -ve}$$

"The point is maximum"

Complex Engineering Problem

Problem:-

A cylindrical water tank with a height of 10 meters and a radius of 5 meters is being filled at a variable rate. The rate at which water is entering the tank is given by the function

$$R(t) = 20 + 4t^2 \text{ (cubic meters / m)}$$

where t is in minutes.

part 1:-

How much water is in the tank after 5 minutes?

part 2:-

At what rate is the water level rising at $t = 5$ minutes, assume the water level is uniform across the surface of the tank.

Background:-

we are tasked with determining how much water is added to a cylindrical tank over time and how quickly the water level rises we need to analyze the amount of water and the rise in the water level at a given time.

Formulation:-

water inflow rate is given by:-

$$R(t) = 20 + 4t^2$$

1. How much water is in the tank by 5min
2. At what rate water level is rising at $t = 5$ min

Data:-

- radius:- 5m
- Inflow rate:-
 $R(t) = 20 + 4t^2, \text{ m}^3/\text{min}$
- Time of interest = 5 minutes

Solutions:-

part 1:- "Total water Volume by 5 minutes"
by integrating the inflow rate at $\int_0^5$ we
find total volume of water.

$$V(5) = \int_0^5 (20 + 4t^2) dt = \left[20t + \frac{4t^3}{3} \right]_0^5$$

$$V(5) = \left[20(5) + \frac{4(5)^3}{3} \right] - [0]$$

$$= 266.67 \text{ m}^3$$

part 2:-

rate of

The volume of

$$V(t) = \pi r^2 h$$

of water at

The rate a

$$\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$$

At $t = 5$

$$\frac{dV}{dt} = R(t)$$

Substituting

$$120 = \pi (5^2) \frac{dh}{dt}$$

$$\frac{dh}{dt} = 1.53$$

part 2:-

rate of water level rise by 5 minutes
The volume of water is related to height by

$V(t) = \pi r^2 h(t)$, where $h(t)$ is the height of water at time 't'

The rate at which water rises is given by:-

$$\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$$

At $t = 5$

Conclusion: After 5 minutes
266.67 m³ of water has
entered, and rising at
approximately 1.53 m/minute.

$$\frac{dV}{dt} = R(5) = 20 + 4(5)^2 = 120 \text{ m}^3/\text{min}$$

Substituting $r = 5$ and solving for $\frac{dh}{dt}$

$$120 = \pi (5^2) \frac{dh}{dt}$$

$$\frac{dh}{dt} = 1.53 \text{ meters/minute}$$

Complex Engineering Problem

- Q A concert promoter is planning an outdoor concert and is considering the layout of the seating. The total area available for seating is 5000 square metres. The promoter estimates that the ticket price can be set at \$50 per ticket, but for every 100 tickets sold the price must be lowered by \$5 to accommodate demand. Maximize the revenue generated from the concert.

Let x be the number of tickets sold in hundreds. Then if $x = 5$ then tickets 500 tickets are sold

$$\text{Price of a ticket} = P = 50 - 5x$$

The total revenue can be expressed

$$R = P \cdot (\text{tickets sold})$$

$$R = (50 - 5x)(100x)$$

$$R = 5000x - 500x^2$$

Finding First derivative:

$$R' = \frac{d}{dx} (5000x - 500x^2)$$

$$R' = 5000 - 1000x$$

Finding Second Derivative:

$$R'' = \frac{d^2}{dx^2} (5000x - 500x^2)$$

$$R'' = (-1000)$$

Since $R'' < 0$, this indicates a maximum

• Solving for x

$$R' = 0 \rightarrow 5000 - 1000x = 0$$

$$5000 = 1000x$$

$$\underline{x = 5}$$

Substitute $x = 5$ into the price of the function to calculate optimal ticket price

$$P = 50 - 5x$$

$$P = 25$$

The total number of tickets sold are $100x \rightarrow 100(5) = 500$

Now calculate the total revenue

$$R = P \cdot \text{Number of tickets sold}$$

$$25 \cdot 500$$

Conclusion:-

To maximize revenue, the promoter should sell 500 tickets at a price of \$25 each, resulting in total revenue of \$12,500

Interpretation:

The concert revenue problem illustrates how ticket prices influence demand, aiming to maximize total revenue within constraints like seating capacity. By using calculus, the promoter can determine the optimal price that balances sales and profitability, applicable to various industries.

Aroosa Azeem
SE - 018

Complex Engineering Problem: Taylor Series

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- In signal processing sinusoidal are often represented using Taylor or Maclaurin Series for approximation. Derive Taylor Series approximation for $\sin x$ up to the 5th term and use it to approximate value of $\sin(1.5)$ Compare this approximation with the actual value. How does the error behave as ^{if degree of} polynomial increases.

Background:

In signal processing, sinusoidal functions such as $\sin x$ are frequently approximated using Taylor Series. This series provide polynomial approximation that can simplify computations especially in digital signal processing. The Mac

Taylor Series Formula:

The Taylor Series expansion of a function $f(x)$ is given by:—

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots$$

Taylor Series Formula:

The Taylor Series expansion of a function $f(x)$ is given by:—

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \dots$$

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CS CamScanner

at $x = 0$

$$f(0) = f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!} + \dots$$

For the sine function's Derivative calculated as:—

$$f(x) = \sin x$$

$$f'(x) = \cos x$$

$$f''(x) = -\sin x$$

$$f'''(x) = -\cos x$$

$$f^{(4)}(x) = \sin x$$

$$f^{(5)}(x) = \cos x$$

Solution:

Step 1 Evaluate Derivatives at $x=0$

$$f(0) = \sin(0) = 0$$

$$f'(0) = \cos(0) = 1$$

$$f''(0) = -\sin(0) = 0$$

$$f'''(0) = -\cos(0) = -1$$

$$f^{(4)}(0) = \sin(0) = 0$$

$$f^{(5)}(0) = \cos(0) = 1$$

Step 2 Construct the Taylor Series.

$$\sin x \approx 0 + \frac{1 \cdot x^1}{1!} + \frac{0 \cdot x^2}{2!} - \frac{1 \cdot x^3}{3!} + \frac{0 \cdot x^4}{4!} + \frac{1 \cdot x^5}{5!}$$

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$$= x - \frac{x^3}{6} + \frac{x^5}{120}$$

$$= x - \frac{x^3}{6} + \frac{x^5}{120}$$

Step:3 For Approximation Substitute 1.5 :-

Now we can approximate $\sin(1.5)$:

$$\sin(1.5) \approx 1.5 - \frac{(1.5)^3}{6} + \frac{(1.5)^5}{120}$$

$$\sin(1.5) \approx 1.5 - \frac{3.375}{6} + \frac{7.59375}{120}$$

$$\sin(1.5) \approx 1.5 - 0.5625 + 0.06328$$

$$\approx 1.00078$$

But, Actual Value of $\sin(1.5)$:

using a calculator, the actual value of $\sin(1.5)$ is approximately 0.997495

Result :-

Step:4 Comparison Of Approximate & Actual Value :

Approximation : 1.00078125

Actual Value : 0.997495

Error :

$$1.00078125 - 0.997495 \approx 0.00328625$$

Conclusion: (Error Behavior)

As the degree of the polynomial increases in Taylor Series, the approximation generally becomes more accurate, especially

for values of x close to the expansion point here ($x=0$). The error decreases as higher-order terms are included, leading to better representation of function.

In this case, although the 5th order approximation was reasonably close, increasing the order further would reduce the error, highlighting the convergent behaviour of Taylor Series

Interpretation:

The approximation for $\sin(1.5)$ demonstrates the utility for Taylor Series in Signal Processing, where precise computation are often necessary. This analysis reinforces the importance of understanding approximation methods, especially when dealing with non linear functions like sine in digital systems.

COMPLEX ENGINEERING PROBLEM

Problem:

A particle moves along a straight line such that its position at time t (in seconds) is given by the function:

$$s(t) = t^3 - 6t^2 + 9t + 2$$

- (i) Determine velocity of the particle at any time t .
- (ii) Calculate time t at which particle comes to rest.
- (iii) Determine acceleration of the particle.
- (iv) Analyze the behaviour of particle at time when it comes to rest. Specifically, evaluate the limit of acceleration as t approaches the time when velocity is zero.

Solution:

Step 1: Find Velocity Function

$$\Rightarrow s(t) = t^3 - 6t^2 + 9t + 2$$

Differentiate w.r.t t .

$$\Rightarrow \frac{d}{dt} [s(t) = t^3 - 6t^2 + 9t + 2]$$

$$\Rightarrow \frac{d}{dt} [s(t)] = \frac{d}{dt} (t^3 - 6t^2 + 9t + 2)$$

$$\Rightarrow v(t) = 3t^2 - 12t + 9 \quad \left[\because \frac{d}{dt} s(t) = v(t) \right]$$

Step 2: Determine when particle comes to rest

In order to find the time when particle come to rest, let $v(t) = 0$

$$\Rightarrow v(t) = 0$$

$$\Rightarrow 3t^2 - 12t + 9 = 0$$

$$\Rightarrow 3(t^2 - 4t + 3) = 0$$

$$\Rightarrow t^2 - 4t + 3 = 0$$

$$\Rightarrow t^2 - 3t - t + 3 = 0$$

$$\Rightarrow t(t-3) - 1(t-3) = 0$$

$$\Rightarrow (t-2)(t-1) = 0$$

either $t = 3$ or $t = 1$

The particle comes to rest at $t = 1$ second and $t = 3$ s.

Step 3: Find acceleration Function.

$$\Rightarrow v(t) = 3t^2 - 12t + 9$$

Differentiate w.r.t t on both sides

$$\Rightarrow \frac{d}{dt} v(t) = \frac{d}{dt} (3t^2 - 12t + 9)$$

By using power law, we get

$$\Rightarrow a(t) = 6t - 12$$

$$[\because \frac{d}{dt} \{v(t)\} = a(t)]$$

Step 4: Analyze Behaviour of Particle when it comes to rest:

Now, we will evaluate the acceleration at times $t = 1$ and $t = 3$.

1) At $t = 1$:

$$\Rightarrow a(1) = 6(1) - 12$$

$$\Rightarrow a(1) = -6$$

2) At $t = 3$:

$$\Rightarrow a(3) = 6(3) - 12$$

$$\Rightarrow a(3) = 6$$

Limit Analysis:

To further analyze the behaviour of acceleration at these points, we can find the limit as t approaches to 1 and 3.

Date:

limit as $t \rightarrow 1$:

$$\Rightarrow \lim_{t \rightarrow 1} a(t) = 6(1) - 12$$

$$\Rightarrow \lim_{t \rightarrow 1} a(t) = -6$$

and $t = 3$ sec

limit as $t \rightarrow 3$:

$$\Rightarrow \lim_{t \rightarrow 3} a(t) = 6(3) - 12$$

$$\Rightarrow \lim_{t \rightarrow 3} a(t) = 6$$

Conclusion:

- (i) The velocity of function is $v(t) = 3t^2 - 12t + 9$
- (ii) The particle comes to rest at $t = 1$ sec and $t = 3$ sec.
- (iii) The acceleration of function is $a(t) = 6t - 12$
- (iv) The acceleration at $t = 1$ is -6 , indicating the particle is ^{decelerating} and the acceleration at $t = 3$ is 6 , indicating the particle is accelerating.